

DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario

Fan Zhang[✉], Zhiyong Wang[✉], Haoting Shen[✉], Bolin Yang[✉], Qianmei Wu[✉] and Kui Ren[✉]
School of Cyber Science and Technology, College of Computer Science and Technology, Zhejiang University[✉]

Alibaba-Zhejiang University Joint Research Institute of Frontier Technologies^{*}
Key Laboratory of Blockchain and Cyberspace Governance of Zhejiang Province[✉]

ABSTRACT

With rapidly increasing demands for cloud computing, Field Programmable Gate Array (FPGA) has become popular in cloud datacenters. Although it improves computing performance through flexible hardware acceleration, new security concerns also come along. For example, unavoidable physical leakage from the Power Distribution Network (PDN) can be utilized by attackers to mount remote Side-Channel Attacks (SCA), such as Correlation Power Attacks (CPA). Remote Fault Attacks (FA) can also be successfully presented by malicious tenants in a cloud multi-tenant scenario, posing a significant threat to legal tenants. There are few hardware-based countermeasures to defeat both remote attacks that aforementioned. In this work, we exploit Time-to-Digital Converter (TDC) and propose a novel defense technique called DARPT (Defense Against Remote Physical attack based on TDC) to protect sensitive information from CPA and FA. Specifically, DARPT produces random clock jitters to reduce possible information leakage through the power side-channel and provides an early warning of FA by constantly monitoring the variation of the voltage drop across PDN. In comparison to the fact that 8k traces are enough for a successful CPA on FPGA without DARPT, our experimental results show that up to 800k traces (100 times) are not enough for the same FPGA protected by DARPT. Meanwhile, the TDC-based voltage monitor presents significant readout changes (by 51.82% or larger) under FA with ring oscillators, demonstrating sufficient sensitivities to voltage-drop-based FA.

1 INTRODUCTION

As more powerful computation capability is required for machine learning, FPGA is increasingly utilized as accelerators in cloud computing centers due to its high parallelization and flexible implementation. Meanwhile, services such as FPGA-as-a-Service (FaaS) and Acceleration-as-a-Service (AaaS) are also provided on many cloud platforms [7]. With such services, users can quickly develop customized IPs to achieve profitability. To improve the utilization of FPGA resources, the concept of multi-tenant is proposed, which shares resources on one FPGA among multiple users [29]. The multi-tenant virtualization renders the user units to be isolated from each other at the logic level. However, it fails to cut down the physical relevance at the electrical level between different users as shown

in [9]. For example, different values and operations in the Programmable Logic (PL) will affect the instantaneous voltage and current in Power Distribution Network (PDN). That is, all users' logic on the same FPGA will have a joint effect on the voltage fluctuations in the PDN, which might be exploited by malicious tenants.

In typical physical attacks such as Side-Channel Attacks (SCA) and Fault Attacks (FA), an attacker can recover the secret information by collecting and exploiting the leakage of the device or injecting the fault through local devices. In general, those attacks using near-field devices may not exist in cloud conditions due to physical isolation. Unfortunately, the relation between voltage fluctuation and user implemented logic in the PDN makes the physical attacks feasible in the multi-tenant scenario. Through the PDN, both SCA and FA can be implemented remotely, which are called remote SCA and remote FA respectively. A malicious user has the authority to submit a bitstream that influences the PDN and collects the voltage variation during the execution process stealthily. Remote SCA on some encryption algorithms such as AES and RSA have been implemented in the cloud successfully [23, 26]. In addition, it is feasible to conduct fault injection attacks on other users sharing the same FPGA by executing high-power-wasting designs [9, 12].

As the mechanism of SCA and FA are different from each other, it is challenging to defend against both with one solution. Common strategies to resist SCA include *masking* and *hiding*. Proposed *hiding* designs include clock misalignment and noise compensation [6]. At the software level, the clock misalignment is usually implemented by inserting some dummy operations into the algorithm. Considering the noise compensation, this approach usually introduces considerable overhead due to the numerous amounts of power compensation units [13]. The countermeasures against FA are dedicated to error detection and threshold implementations [3]. The traditional way to detect fault injection leverages a redundant computing unit, which consumes double resources.

In this paper, we propose a novel countermeasure called DARPT which can resist both SCA and FA with relatively low overhead. DARPT is composed of only basic primitives and hence it can be migrated to diverse programmable logic parts for security consideration. Our design mainly consists of a generator of clock jitters and a detector of fault injection for defense against SCA and FA, respectively. The threshold detection method against fault injection with TDC was performed by [16]. Subsequently, we propose a doubled clock frequency generator based on TDC which further boosts the SCA security. Our contributions are concluded as follows:

- We propose a new hardware-based countermeasure to defend against both SCA and FA. The proposed defense exploits the

^{*}Haoting Shen is the Corresponding Author (Email: htshen@zju.edu.cn)

feature of TDC and utilizes it to generate clock jitters to mitigate SCA. To the best of our knowledge, this is the first time that TDC is used as a comprehensive solution to defend both attacks, especially under the multi-tenant scenario of cloud FPGA.

- We show that our design can be implemented on FPGA with low overhead. The countermeasure can be implemented with merely 0.61% LUT and 0.24% FF of onboard resources, not requiring any extra resources such as PLL and BRAM.
- We show that the proposed DARPT against SCA and FA is highly secure. While a successful SCA requires 8000 traces on an FPGA without protection, the same attack will fail even given 100× more traces when DARPT is deployed. Meanwhile, our design can effectively capture suspicious voltage drops resulting from FA.

2 PRELIMINARIES

In this section, we will briefly introduce the state-of-the-art remote SCA and remote FA. The principles of Test Vector Leakage Assessment (TVLA) and Correlation Power Analysis (CPA) used for the evaluation will also be described.

2.1 Remote SCA

For remote SCA performed in the cloud, a typical threat model assumes that the adversary has access to partial Programmable Logic (PL) in the FPGA where the victim implements its design. With a shared Power Distribution Network (PDN), the voltage variation contains information about running operation and medium values. As voltage affects path delays, adversaries can utilize appropriate sensors to monitor voltage variations and further extract sensitive information. In 2018, Schellenberg *et al.* [22] proposed a remote power analysis model to attack Advanced Encryption Standard (AES) in a multi-tenant scenario. The authors implemented Time-to-Digital Converter (TDC) and AES in one FPGA without any logic connections. The TDC worked as a monitor to collect the leakage power information. Through CPA, they successfully recover the secret key by using only 5000 traces. Ring Oscillator (RO) is also implemented as remote sensors due to its sensitivity to frequency changes. For example, Zhao *et al.* [26] tested RO for remote SCA and successfully achieved Simple Power Analysis (SPA) against RSA.

Remote sensors can also be utilized to extract sensitive data from recently surging neural network applications [18]. Higher computational loads in neural networks require higher power consumption, where significant information such as the weights of neurons and the input of neural networks may be leaked through side channels. For example, Moini *et al.* [18] successfully recovered the key message from a trained BNN for MNIST through a TDC-based remote sensor.

2.2 Remote FA

FA is another popular method reported for remote attacking [9, 12]. In [9], a PDN-based DoS attack was presented in a multi-tenant scene by employing a high-speed flipping RO to induce a massive voltage drop. It is shown that the RO could shut down or restart the entire FPGA. Follow-up investigations showed that it is possible to introduce fault injection attacks to a remote FPGA by altering

the number and the frequency of flip RO units. Krautter *et al.* [12] utilized this method to recover the key of an AES module with Differential Fault Attack (DFA) in a multi-tenant setting. Later Luo *et al.* [17] discovered that the remote fault injection can also be used to interfere with neural networks, where RO units implemented in the cloud consumed 15.3% of the PL resources and reduced the model recognition success rate by up to 14% (on an MNIST-trained Lenet-5 network).

2.3 Test Vector Leakage Assessment

To evaluate the potential leakage risk of an algorithm, TVLA has been proposed as an effective assessment [28]. Testers utilize Welch T-tests to detect any possible leakage. The result of this assessment is T value, which is computed as $T = (X_A - X_B) / \sqrt{\frac{S_A^2}{N_A} + \frac{S_B^2}{N_B}}$, where X_A , S_A , and N_A are the mean, standard deviation, and cardinality of group A, respectively. Group A is a set of stationary plaintexts, while group B is a set of varying plaintexts. The test fails if the T value crosses the threshold of $[-4.5, 4.5]$ which shows the algorithm is possibly vulnerable to SCA.

2.4 Correlation Power Analysis

Since CPA was proposed in [8], it has been widely used for SCA due to its effectiveness. To perform CPA, it is necessary to establish a correlation model between the medium values and the instantaneous power consumption. The Pearson's Correlation Coefficient shown in Eq. (1) is considered an effective way for measuring the relevance.

$$\rho_{X,Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{E[(X - \mu_X)^2]E[(Y - \mu_Y)^2]}} \quad (1)$$

Where X and Y are the medium values and instant value of the power consumption, respectively. $E[(X - \mu_X)(Y - \mu_Y)]$ is the covariance of X and Y , and the denominator is composed of the standard deviation of X and Y .

Considering the structure of FPGA, the hamming distance which stands for the number of bit flips is usually confirmed as the medium value for calculation.

3 RELATED WORKS

In this section, we conclude some countermeasures against SCA and FA in the multi-tenant situation, including the ones against either SCA or FA and the ones against both.

As mentioned in Sec. 1, hiding sensitive signals is an efficient strategy to prevent SCA. Pammu proposed a circuit containing a state-shift clock to misalign the traces [19]. Considering the software implementation, [27] achieved clock misalignment by randomly repeating the genuine operations with dummy values. [4] proposed a novel defense against SCA which hiding the correct key by providing a fake key through statistical methods. Krautter *et al.* [13] presented a novel active fence to hide possible leakage by decreasing the Signal-to-Noise Ratio (SNR). [2] proposed the first fully-masked neural network inference engine design against side-channel attacks, which maintains the safety of the system with up to 2M traces.

The FA countermeasures can be categorized into *threshold alarm* and *error detection*. Haarman combined both to produce the ParTI

countermeasure for FA [3]. Considering the defense in design level, another novel framework named SOLOMON was proposed to detect underlying fault attacks by checking the bitstream [24]. In [16], a dynamic frequency scale framework was proposed to impede fault attacks. They utilized a carry-chain delay line to detect the voltage-drop attacks. For the security in the cloud environment, George utilized onboard sensors to help Cloud Service Provider (CSP) accurately locate the destructive combinational loop units and remove them from FPGA's PL part [20]. As Amazon Web Services disables composition loop logic in the cloud [1], many loop units are proposed with mixing sequential elements and other basic primitives to avoid Design Rule Check (DRC) [11]. Therefore, FPGADefender [15] was introduced to solve possible loop units by parsing the implementation graph and parsing scanning options to seek malicious circuits and constructs.

Meanwhile, researchers also proposed countermeasures against both SCA and FA. For instance, Gnad and Rapp introduced one bitstream checking tool to protect shared FPGAs from SCA and FA [10]. KRAUTTER formulated several comprehensive attack signatures for the attack detection in user bitstream during checking mapped netlists [14]. Bhasin and Ravi proposed PLL-based Protection Against Physical attack (PPAP) and the improved method (iPPAP) to secure IP core from SCA [21]. An internal RO was used to detect FA in their design. Yao proposed a combined countermeasure against SCA and FA named PRO [25]. PRO utilized the frequency sensitivity of RO to perform threshold detection for FA and introduced RO cells to compensate for the power consumption generated by the core.

Table 1: Related Works of Countermeasure against Remote Physical Attack

Attack Type	Defense Method	Implement Type	Refs.
SCA	Hiding	Hardware	[4],[13]
	Clock Misalignment	Hardware	[19]
	Masking	Hardware	[2]
	Clock Misalignment	Software	[27]
FA	Bitstream Checking	Software	[11],[15],[24]
	Threshold Detecting	Hardware	[3],[16],[20]
SCA&FA	Bitstream Checking	Software	[10],[14]
	Clock Misalignment & Threshold Detecting	Hardware	[21],[25] Our work

To summarize different attacks and the corresponding countermeasures, we categorize recent related works and summarize the adopted defense techniques and the implementation manners in Table 1. There are some limitations with current solutions. For example, cloud service providers are considered untrusted in many cases. The uploaded bitstream is encrypted for security reasons and IP protection, which hinders the netlists checking and DRC and disables direct bitstream checking [10, 14]. In [21, 25], the defense based on RO structure has problems for the implementation in cloud platforms, since the combinatorial loops in designs are not supported by CSP [1].

4 TDC-BASED COUNTERMEASURE

As described in Sec. 1, TDC can generate delayed signals helping hide the sensitive message in power variations and alert to voltage drop. Therefore, TDC is selected as the core component in our

DARPT design. The design is expected to ensure the security under both SCA and FA. Then we will introduce the basic structure of TDC and the design of DARPT.

4.1 Basic Structure

The basic structure of TDC in FPGA is illustrated in Fig. 1. It consists of delay units and D flip-flops. The number of delay units passed by the *Hit* signal in a clock cycle reflects the voltage magnitude of the PDN. The fluctuation of voltage can be attributed to the inductive and resistive components in FPGA, described by $V_{drop} = L \cdot \frac{dI}{dt} + IR$. Thus, the current in PDN has a significant effect on the voltage. Different operations lead to current changes, resulting in voltage fluctuations, which may affect the circuit delay and the output of TDC, as well.

In general, the *Hit* signal is provided by the initial delay line which creates sufficient phase deviation. Another optional way is to directly include the *Hit* signal with a shifted phase. The temporary value of the flip-flops Q_x constitutes the output of TDC. Through a decoder, the quantitative voltage amplitude fluctuation can be obtained. The sampling accuracy of the TDC is determined by the time delay of a single unit and the total number of delay units.

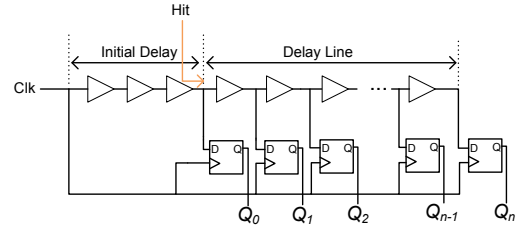


Fig. 1. TDC based on tapped delay line

The output of the TDC is N , which is determined by $N = \frac{T_{delay}}{t_{unit}}$. T_{delay} is the time needed to pass the delay line, t_{unit} is determined by the selection of the delay unit, the instantaneous supply power and the processing technology of transistors. Usually, t_{unit} can be regarded as a constant value.

4.2 Our Design

The proposed TDC-based countermeasure is illustrated in Fig. 2. The delay unit contained in the design is implemented with carry-chains, which is one primitive of FPGA. The latches are composed of 256 flip-flops which can be easily managed with FPGA on-chip resources, such as FDCE and FDR. The multiplexer and Pseudo Random Number Generator (PRNG) are implemented with Look-Up-Table (LUT). In a standard FPGA, the design can be performed with PL resources easily.

In our design, we use the signal *CLK* followed by a 180° phase shift as the *Hit* signal, as shown in Fig. 1. The *output*[255 : 0] of latches provides real-time voltage information of the whole network. With the *output* signal, the PRNG module produces the *SEL*[7 : 0] for the multiplexer through simple XOR operations. The output of latches and the *PresetThreshold* are compared to detect abnormal changes in PDN. The *Output_clk* is the input clock for the core protection and the *Alarm* signal is activated when a massive voltage change is detected.

The delay line structure makes TDC sensitive to voltage change, which can be easily utilized as a threshold countermeasure against

frequency of the ROs is 25MHz. We distribute all power consumption units into 16 fractions, each containing 512 ROs. The control signals of these fractions are separated. In this way, we manage the RO fractions in a flexible manner to generate different degrees of voltage oscillation.

5.2 Evaluation of SCA Prevention

In this section, we leverage the leakage behavior assessment tool (TVLA) and practical side-channel attacks (CPA) to evaluate the security performance of our TDC-based countermeasure against SCA. We take the typical hardware-implemented AES-128 as an example. It is worth noting that the output CLK signal from DARPT serves as the input clock signal for AES. We utilize a 256-bit TDC as a deployable sensor to achieve the remote SCA. For comparison, both unprotected and protected AES modules are the target of our evaluation.

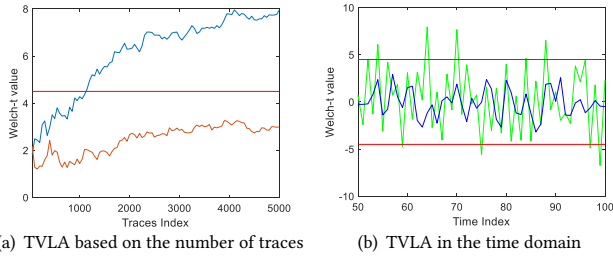


Fig. 5. a) The brown line is for the protected AES module and the cyan-blue line is for the unprotected AES module. b) The blue line and the green line represent the AES core with and without defense, respectively. The red line indicates the TVLA threshold of ± 4.5 for both pictures.

In Fig. 5(a), Welch's value represents the maximum t value of sample point from 50 to 100. It can be seen that when the number of traces is more than 1400, leakage is likely to occur on the unprotected AES (indicated by $t \geq 4.5$), and keeps increasing with the increasing number of traces. However, from the AES protected by DARPT, Welch's value is well controlled until the test reaches the maximum of 5000 traces. Fig. 5(b) illustrates the TVLA results in the time domain. We can see that the t value of protected AES is maintained below 4.5, while the unprotected one indicates leaking on some points. Overall, the leakage on FPGA deployed with DARPT is significantly reduced, practically showing improved security against SCA.

Though TVLA is an effective tool to assess the leakage behavior of an algorithm [28], whether these leakages can be eventually exploited by adversaries to recover the secrets is not guaranteed from TVLA results. Therefore, we further perform the CPA to evaluate the security improvement made by DARPT. For both unprotected and protected AES, we collect 10,000 traces from each one. The working frequency of AES core and the sampling rate of TDC follow the setting described in Sec. 5.1.

The CPA results from unprotected and protected AES modules are shown in Fig. 6. All attack targets are in the last round of AES. Fig. 6(a) shows that the correct key (highlighted in red) of unprotected AES can be successfully extracted within 10,000 traces. While from Fig. 6(b), it shows that DARPT can protect the correct

key (highlighted in blue) from being recovered. The most possible key candidate (highlighted in red) turns out to be an incorrect key guess. To further illustrate the security difference between unprotected and protected AES, we complement more traces analysis and take the Minimum Traces for Key recovery (MTK) as the metric, with results shown in Fig. 6(c) and (d), where the red line indicates the correct keys. In Fig. 6(c), we can see that the AES module without any protection exposes vulnerabilities as MTK is only about 8,000 traces, sharing the same order of magnitude with the CPA using remote sensors [22, 23]. Contrary to the unprotected one, when the AES is protected by DARPT, we do not find any significant evidence of correct key extraction until the analysis reaches the maximum of 800,000 traces in our experiment. This can be attributed to the reason that the selected signal is randomized by PRNG, which makes the information of delay difficult to be leveraged. In conclusion, CPA results demonstrate that DARPT significantly improves the security performance of FPGA instead of perfect defense SCA in multi-tenant scenarios.

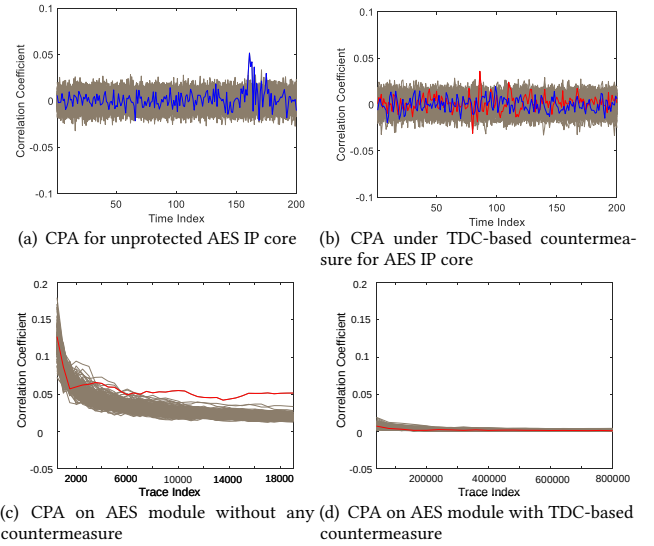


Fig. 6. a) The highlighted blue line stands for the right tenth round key which is revealed in CPA and b) the highlighted red line and blue line represent the guessed value and the real tenth round key, respectively. c) shows the MTK for AES module without countermeasure and d) gives the MTK for AES module under TDC-based countermeasure.

5.3 Evaluation of FA Detection

Due to the limited user permission in cloud multi-tenant conditions, precise fault injection on a particular location is not achievable. Unlike [20], the location of fault injection is not critical in our evaluation. Instead, we focus on the reliability of detection for all possible FA that are based on voltage or temperature fluctuations.

Thanks to the stable power supply system on modern FPGA, a reliable baseline for DARPT monitoring is available the most time. Once the baseline is built, DARPT can effectively detect abnormal voltage changes across PDN. By selecting different numbers of control signals of RO fractions, different degrees of voltage fluctuation can be generated. As shown in Table 3, more RO units enabled at

the same oscillating frequency lead to a more voltage variance in the PDN. When all RO units are activated, the voltage fluctuation increases to about five times the normal level. A proper range limitation can be preset for the alarm signal, based on the analysis of the normal sample traces and the traces under remote fault attacks.

Table 3: Relationship of RO fractions and Fluctuation of TDC sampling values

Number of enabled RO fractions	TDC value change ratio (compared with normal value)(%)
1	51.82
2	75.30
4	119.03
8	225.91
16	459.92

When the PDN fluctuation in the FPGA caused by attackers or other abnormal conditions exceeds the predefined limits, it is detected and the alarm signal is triggered. The defense module shall be aware of the abnormal power supply network behavior on the FPGA for the first time. It then calls error handlers and executes corresponding protective activities for the operation they are processing, to avoid any potential detrimental consequences.

6 CONCLUSION

In this work, we develop a novel DARPT design to protect FPGA from remote SCA and FA in a multi-tenant cloud scenario. The DARPT is based on primitive designs, making it applicable to various FPGAs with low overhead. The design exploits doubled clock frequency technic to boost SCA security by misaligning the clock. Theoretical calculation (TVLA) and experimental attacking (CPA) are performed for security evaluation, shown that DARPT is efficient to prevent SCA. Meanwhile, the TDC in DARPT demonstrates sufficient sensitivities to detect voltage-drop-based FA. Further researches focus on evaluating the defense effect for SCA in the frequency domain and the resistance against resynchronization.

ACKNOWLEDGMENTS

This work was supported in part by National Key R&D Program of China (2020AAA0107700), by National Natural Science Foundation of China (62072398, 62032021, 61772236), by Alibaba-Zhejiang University Joint Institute of Frontier Technologies, by National Key Laboratory of Science and Technology on Information System Security, by State Key Laboratory of Mathematical Engineering and Advanced Computing, and by Key Laboratory of Cyberspace Situation Awareness of Henan Province.

REFERENCES

- [1] AWS. 2021. Amazon Web Services. Retrieved 2021-10-30 from <https://github.com/aws/aws-fpga/blob/master/ERRATA.md>
- [2] Anuj Dubey, Rosario Cammarota, and Aydin Aysu. 2020. BoMaNet: Boolean masking of an entire neural network. In *2020 IEEE/ACM International Conference On Computer Aided Design (ICCAD)*. IEEE, 1–9.
- [3] TR Haarman. 2020. *Analysing countermeasures against fault injection attacks on FPGA based cryptographic implementations*. B.S. thesis. University of Twente.
- [4] Ruben Lumbiarres-Lopez, Mariano Lopez-Garcia, and Enrique Canto-Navarro. 2016. Hardware architecture implemented on FPGA for protecting cryptographic keys against side-channel attacks. *IEEE Transactions on Dependable and Secure Computing* 15, 5 (2016), 898–905.
- [5] Dina Mahmoud and Mirjana Stojilović. 2019. Timing violation induced faults in multi-tenant FPGAs. In *2019 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 1745–1750.
- [6] Maamar Ouladj and Sylvain Guilley. 2021. *Side-Channel Analysis of Embedded Systems: An Efficient Algorithmic Approach*. Springer Nature.
- [7] Amazon Web Services. 2021. Amazon EC2 F1 instances. Retrieved 2021-10-30 from <https://aws.amazon.com/cn/ec2/instance-types/f1/>
- [8] Brier et al. 2004. Correlation power analysis with a leakage model. In *International workshop on cryptographic hardware and embedded systems*. Springer, 16–29.
- [9] Gnad et al. 2017. Voltage drop-based fault attacks on FPGAs using valid bitstreams. In *2017 27th International Conference on Field Programmable Logic and Applications (FPL)*. IEEE, 1–7.
- [10] Gnad et al. 2018. Checking for electrical level security threats in bitstreams for multi-tenant FPGAs. In *2018 International Conference on Field-Programmable Technology (FPT)*. IEEE, 286–289.
- [11] Giechaskiel et al. 2019. Measuring long wire leakage with ring oscillators in cloud FPGAs. In *2019 29th International Conference on Field Programmable Logic and Applications (FPL)*. IEEE, 45–50.
- [12] Krautter et al. 2018. FPGAhammer: Remote voltage fault attacks on shared FPGAs, suitable for DFA on AES. *IACR Transactions on Cryptographic Hardware and Embedded Systems* (2018), 44–68.
- [13] Krautter et al. 2019. Active fences against voltage-based side channels in multi-tenant FPGAs. In *2019 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 1–8.
- [14] Krautter et al. 2019. Mitigating electrical-level attacks towards secure multi-tenant FPGAs in the cloud. *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 12, 3 (2019), 1–26.
- [15] La et al. 2020. FPGADefender: Malicious Self-Oscillator Scanning for Xilinx UltraScale+ FPGAs. *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 13, 3 (2020), 1–31.
- [16] Luo et al. 2020. A quantitative defense framework against power attacks on multi-tenant FPGA. In *2020 IEEE/ACM International Conference On Computer Aided Design (ICCAD)*. IEEE, 1–4.
- [17] Luo et al. 2021. DeepStrike: Remotely-Guided Fault Injection Attacks on DNN Accelerator in Cloud-FPGA. *arXiv preprint arXiv:2105.09453* (2021).
- [18] Moini et al. 2021. Remote power side-channel attacks on BNN accelerators in FPGAs. In *Design, Automation & Test in Europe (DATE)*.
- [19] Pammum et al. 2017. Highly secured state-shift local clock circuit to countermeasure against side channel attack. In *2017 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 1–4.
- [20] Provelengios et al. 2019. Characterizing power distribution attacks in multi-user FPGA environments. In *2019 29th International Conference on Field Programmable Logic and Applications (FPL)*. IEEE, 194–201.
- [21] Ravi et al. 2018. Ppap and ippap: PLL-based protection against physical attacks. In *2018 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*. IEEE, 620–625.
- [22] Schellenberg et al. 2018. An inside job: Remote power analysis attacks on FPGAs. In *2018 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 1111–1116.
- [23] Schellenberg et al. 2018. Remote inter-chip power analysis side-channel attacks at board-level. In *2018 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 1–7.
- [24] Srivastava et al. 2020. SOLOMON: An automated framework for detecting fault attack vulnerabilities in hardware. In *2020 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 310–313.
- [25] Yao et al. 2021. Programmable RO (PRO): A Multipurpose Countermeasure against Side-channel and Fault Injection Attack. *arXiv preprint arXiv:2106.13784* (2021).
- [26] Zhao et al. 2018. FPGA-based remote power side-channel attacks. In *2018 IEEE Symposium on Security and Privacy (SP)*. IEEE, 229–244.
- [27] Stefan Tillich, Christoph Herbst, and Stefan Mangard. 2007. Protecting AES software implementations on 32-bit processors against power analysis. In *International Conference on Applied Cryptography and Network Security*. Springer, 141–157.
- [28] TVLA. 2021. Test Vector Leakage Assessment (TVLA) Derived Test Requirements (DTR) with AES. Retrieved 2021-10-30 from <https://www.rambus.com/wp-content/uploads/2015/08/TVLA-DTR-with-AES.pdf>
- [29] Yue Zha and Jing Li. 2020. Virtualizing fpgas in the cloud. In *Proceedings of the Twenty-Fifth International Conference on Architectural Support for Programming Languages and Operating Systems*. 845–858.